

NARAMREDDY CHARAN KUMAR REDDY

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PROFESSIONAL SUMMARY

3rd-year Computer Science student specializing in AI and Machine Learning, building end-to-end systems that reach production rather than stopping at a notebook metric. Experience spans medical-imaging deep learning, gradient-boosted forecasting across thousands of parallel series, and recommendation architectures on learned embeddings — plus the serving layer around them. Strong bias toward leakage-free validation and metrics that describe real behaviour.

TECHNICAL SKILLS

Machine Learning:	XGBoost, LightGBM, Random Forest, Logistic Regression, Scikit-Learn, Optuna.
Deep Learning & CV:	PyTorch, Transfer Learning (ResNet / DenseNet / EfficientNet), CNNs, pydicom, Neural Collaborative Filtering, Vector Embeddings, FAISS.
Data Intelligence:	Pandas, NumPy, Feature Engineering, KNNImputer, Imbalance Handling, RobustScaler.
Engineering & Deploy:	Python 3 (Advanced OOP), Data Structures, FastAPI, REST APIs, SQL (Schema Design), SQLAlchemy, Docker, Azure Container Apps, Git, Claude Code (agentic development).

ENGINEERING PROJECTS

RSNA Pneumonia Detection | Chest X-Ray Screening | PyTorch, pydicom, FastAPI, [NaramCharan/RSNA-Pneumonia-Detection](https://naramcharan.github.io/RSNA-Pneumonia-Detection)
React, Docker, Azure

Live application: rsna-app.salmonmeadow-7644e67e.eastasia.azurecontainerapps.io

- Benchmarked DenseNet-121, EfficientNet-B2 and ResNet-34 under an identical two-stage transfer-learning protocol; the fine-tuned ResNet-34 reached **83% pneumonia recall, 0.79 F1 and 86% accuracy** on 1,836 held-out studies, against a 68% always-negative baseline.
- Diagnosed a RandomCrop(224) transform discarding 95% of every 1024×1024 radiograph while retaining the study-level label — removing it produced a larger gain than every architecture change in the project combined.
- Prevented patient-level leakage by splitting on unique patient IDs rather than rows, evaluating the test set once; deployed as one Docker container on Azure serving a React interface and REST API, cutting hosting cost 84%.

Walmart Store Weekly Sales Forecasting | Python, Scikit-Learn, XGBoost, [NaramCharan/Walmart-Store-Weekly-Sales-Forecasting](https://naramcharan.github.io/Walmart-Store-Weekly-Sales-Forecasting)
LightGBM, Pandas

- Engineered a global multi-series forecasting pipeline predicting weekly sales across ~3,000 store-department combinations, achieving **95.55% validation R²** with an optimized LightGBM regressor.
- Built a custom recursive walk-forward inference system from scratch to project rolling and lag features over future validation horizons where actual sales are missing.
- Eliminated temporal data leakage by splitting before feature extraction and computing lag/rolling statistics strictly within each partition.

E-Commerce Customer Churn Prediction | Python, Scikit-Learn, XGBoost, [NaramCharan/e-commerce-customer-churn-prediction](https://naramcharan.github.io/e-commerce-customer-churn-prediction)
PyTorch, Optuna

- Tuned tree classifiers with Optuna over 10-fold Stratified Cross-Validation, achieving **98.28% accuracy and 94.74% F1** on the final selected XGBoost production model; a 4-hidden-layer PyTorch network scored 98.76% and was rejected as not worth the compute.

Handbuilt Transformer | Learning project | PyTorch, NLP [NaramCharan/Handbuilt-Transformer](https://naramcharan.github.io/Handbuilt-Transformer)

- Built a mini GPT and a mini GPT tokenizer from scratch in PyTorch to learn how ChatGPT-style models work: a ~0.21M-parameter decoder-only Transformer (causal multi-head attention, 4 blocks) trained on Shakespeare, and a GPT-2-style byte-level BPE tokenizer with regex chunking, checked against **tiktoken**; ongoing — next, training on BPE tokens.

Personal Portfolio — naramcharan.me | Next.js 16, React 19, TypeScript, Three.js, [NaramCharan/NaramCharan.github.io](https://naramcharan.github.io)
Tailwind v4, GitHub Actions

Live site: naramcharan.me

- Shipped a WebGL-hero portfolio with interactive project briefs, statically exported and auto-deployed to GitHub Pages; built by **directing Claude Code** across 75+ commits (architecture and product decisions mine, every change reviewed), lifting Lighthouse from 65 to 87.

EDUCATION

B.Tech in Computer Science Engineering (AI & Machine Learning) — 3rd Year Graduation: May 2028
GD Goenka University — Gurugram, India Current CGPA: 8.98 / 10.0

CERTIFICATIONS

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| • Deep Learning Specialization – DeepLearning.AI (Jul 2026) | • Claude Code: Agentic Coding Assistant – DeepLearning.AI, Anthropic (Jul 2026) |
| • Machine Learning Specialization – DeepLearning.AI, Stanford (May 2026) | • Databases and SQL for Data Science – IBM, Coursera (Nov 2025) |
| • Python for Everybody – Univ. of Michigan (Mar 2025) | • Prompt Engineering & GenAI – Google, Vanderbilt (Mar 2025) |